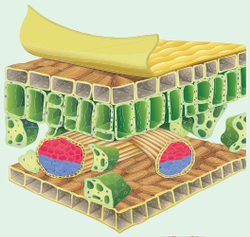
The Plumbing system of Plants and Humans





Plants and animals evolved separately but developed comparable structures to control transport of water and dissolved chemicals. But whose transport system is optimally designed to offer selective advantage? In plants, transport through xylem has allowed growth in height and colonization of diverse habitats and the system has to be extensive as Photosynthesis requires water. Murray's law predicts the thickness of branches in transport networks, such that the cost for transport and maintenance of the transport medium is minimized. This law is observed in the vascular and respiratory systems of animals, xylem in plants, and the respiratory system of insects. Further research in this area will improve our understanding of natural world.

i. **Short-distance (Cell to cell transport):** Involvement of few cells, mostly in the lateral direction. They are the connecting link to xylem and phloem from root hairs or leaf tissues respectively. Examples: Diffusion, Imbibition, and Osmosis.

ii. **Long-distance transport:** Transport within the network of xylem or phloem is an example for long-distance transport. Examples: Ascent of Sap and Translocation of Solutes.

❖ Based on energy expenditure during transport, they are classified as **(a) passive transport and (b) active transport.**

i. **Passive transport:** It is a downhill process which utilizes physical forces like gravity and concentration. No energy expenditure is required. It includes diffusion, facilitated diffusion, imbibition, and osmosis.

ii. **Active transport:** It is a biological process and it runs based on the energy obtained from respiration. It is an uphill process.

11.2 Cell to Cell Transport

Cell to cell or short distance transport covers the limited area and consists of few cells. They are the facilitators or tributaries to the long-distance transport. The driving force for the cell to cell transport can be passive or active (Figure 11.1). The following chart illustrate the various types of cell to cell transport:

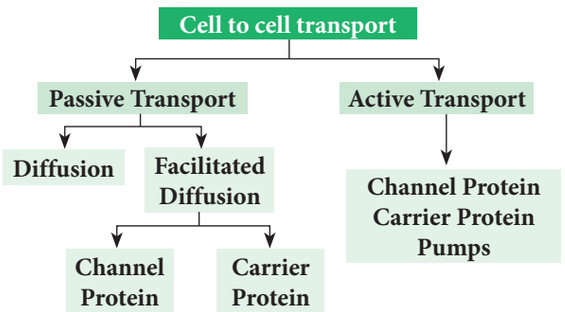


Figure 11.1: Cell to cell transport

11.2.1 Passive Transport

1. Diffusion

When we expose a lightened incense stick or mosquito coil or open a perfume bottle in a closed room, we can smell the odour everywhere in the room. This is due to the even distribution of perfume molecules throughout the room. This process is called **diffusion**.

In **diffusion**, the movement of molecules is continuous and random in order in all directions (Figure 11.2).

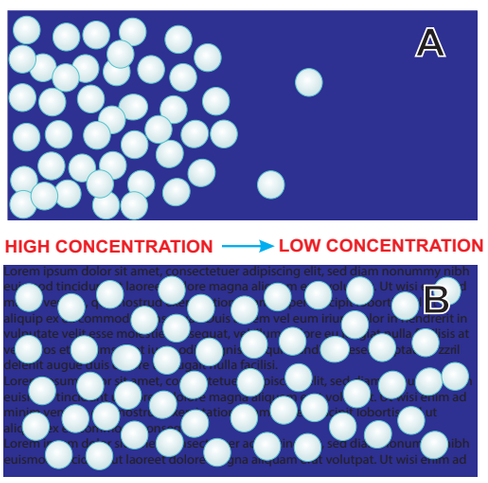


Figure 11.2: Distribution of molecules in diffusion (A) Initial stage (B) Final stage

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